

Project Proposal

ECE 57000: Artificial Intelligence

Summer 2026

1. Overview

The course project is a major component of this class, allowing you to explore a specific AI topic in depth over the semester. This proposal is your first project milestone. You must choose a project track, formulate a clear problem, review prior work, and establish how you will evaluate your success.

The Three Project Tracks

Before beginning your proposal, you must select one of the following tracks for your term project:

- **Tiny Reproductions:** Distill and verify the core claims of a significant research paper using scaled-down models and datasets.
- **Product Prototype:** Engineer a functional AI-powered prototype to solve a specific, real-world user problem.
- **Novel Research:** Identify a research gap, propose a novel hypothesis or method, and rigorously test it for a workshop-level paper.

2. Submission Format

You must submit your proposal as plain text in **Quarto RevealJS markdown format**. Your submission must contain exactly **five slides**. You may use advanced RevealJS markdown (like two-column layouts) if helpful.

If you are unfamiliar with this presentation format, please review the official Quarto RevealJS Basics to get started and understand the syntax.

Constraint: There is a strict **7500-character limit**. Any content exceeding this limit will be truncated during grading.

3. Content Requirements

Slide 1: Track Selection & Problem Formulation

Clearly state which of the three tracks you have chosen. Then, define a concrete problem that you can make meaningful progress on within the course timeline.

- **Well-defined:** The problem statement should be clear and specific, avoiding vague language.
- **Insightful:** Explain why this matters. If you succeed, what difference will it make?
- **Feasible:** Ensure it does not require more time or compute than is reasonable for the semester.

Slides 2–4: Literature Review (One Paper per Slide)

Identify three research papers from peer-reviewed venues (e.g., NeurIPS, ICML, ICLR, CVPR) or arXiv. Do not cite websites or news articles. You may include at most one survey paper.

For each paper, you must:

- Briefly summarize the main contribution of the paper.
- Briefly explain how this specific paper informs or relates to your proposed project.

Slide 5: Evaluation Metric

Propose a metric that objectively and accurately evaluates approaches to your proposed problem.

- **Propose & Justify:** The metric must be computable and quantifiable. Explain why performing well on this metric implies your problem is effectively solved.
- **Contextualize (The 4 Tiers):** Define and justify what ranges of your metric constitute "Poor," "Okay," "Good," and "Excellent" performance in the specific context of your problem (e.g., 51% accuracy is excellent for predicting stock prices, but poor for diagnosing a disease with a 90% baseline).

4. Grading Rubric

Your proposal will be evaluated according to the three equally weighted criteria below.

Problem Formulation

- **Excellent (5):** The problem is exceptionally clear, insightful, and feasible. Represents a meaningful step toward a larger goal.
- **Good (4):** The problem is well-defined and feasible. It is a reasonable step toward a known goal.
- **Satisfactory (3):** The problem is proposed but lacks some clarity, insight, or feasibility.
- **Okay (2):** The problem is too broad, unfeasible, or lacks a clear purpose.
- **Poor (1):** The problem is trivial, nonsensical, or missing entirely.

Literature Review

- **Excellent (5):** Cites 3 appropriate papers. Demonstrates an excellent, concise understanding of each contribution and its relevance.
- **Good (4):** Cites 3 appropriate papers. Summaries and relevance are solid and well-reasoned.
- **Satisfactory (3):** Cites 3 papers, but some may be from weak venues, or the connection to the project lacks depth.
- **Okay (2):** Literature review is superficial. Papers may be missing or from inappropriate sources (e.g., blogs).
- **Poor (1):** No meaningful analysis. Papers are unrelated to the problem, or no papers are cited.

Evaluation Plan

- **Excellent (5):** Proposes a quantifiable, well-justified metric uniquely suited to the problem. The 4 performance tiers are rigorously defined.

- **Good (4):** Proposes a computable metric with a clear connection to the problem. Tiers are defined with reasonable justification.
- **Satisfactory (3):** A metric is proposed and is generally accurate but lacks strong justification. Tiers are present but poorly justified.
- **Okay (2):** The proposed metric is vaguely defined, or the required performance tiers are missing.
- **Poor (1):** The metric is inaccurate, not quantifiable, or missing entirely.